Groundwater Levels for Environmental Sustainability

Course: Concept & technology of AI

Assignment: Assignment 03

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1. Introduction

1.1 Problem Statement

Water is our most precious resource, yet groundwater levels are notoriously difficult to monitor in real-time across vast geographic areas. The goal of this project was to move from reactive to proactive management. By building a regression model, we aim to predict Water Surface Elevation (WSE) accurately, providing a tool for environmental scientists to anticipate shortages before they become crises.

1.2 The Dataset

We worked with the gwl-monthly.csv dataset, a robust collection of over 81,000 monitoring records. It tracks elevation markers such as the Reference Point (RPE) and Ground Surface (GSE), which serve as the skeletal structure for our predictions.

1.3 Why It Matters (UNSDG)

This work isn't just a technical exercise; it aligns directly with UN Sustainable Development Goal 6. By improving our ability to monitor groundwater, we contribute to the global mission of ensuring sustainable water management and sanitation for all.

2. Methodology

2.1 Preparing the Data

Real-world data is rarely "ready-to-use." My first step was ensuring the integrity of the target variable, WSE. A critical discovery during this phase was the presence of potential "data leakage." I identified that certain columns were derived mathematically from our target; keeping these would have given the model the "answers" beforehand, resulting in high but fake accuracy. These were removed to ensure the model actually learned the underlying physical patterns.

2.2 Exploratory Data Analysis (EDA)

Before modeling, I wanted to "see" the data. Using correlation heatmaps, I confirmed a strong relationship between ground elevation and the water table. This confirmed that our features were physically relevant and not just random noise.

2.3 Feature Engineering

Dates were transformed from simple strings into meaningful temporal features like month and year. This allowed the model to account for seasonal fluctuations—essential for water levels that change with the weather.

2.4 Modeling Strategy

I took a two-pronged approach:

1. Task 1 (Neural Networks): I designed a Multi-Layer Perceptron (MLP) with three hidden layers. This allowed the system to capture complex, non-linear relationships that a standard equation might miss.
2. Task 2 (Classical ML): I tested four distinct algorithms—Linear Regression, Ridge, Random Forest, and Gradient Boosting—to see which "logic" best suited the groundwater data.

2.5 Optimization

I didn't settle for "out-of-the-box" settings. Using GridSearchCV, I systematically tested different configurations for the Random Forest model to find the "sweet spot" for accuracy without overfitting.

3. Results and Conclusion

3.1 Key Findings

The results were compelling. While all models performed well, the Neural Network (MLP) and Ridge Regression stood out. The Neural Network, in particular, showed an incredible ability to map the elevation features to the target with minimal error.

3.2 The Final Choice

The Neural Network (MLP) was selected as the champion model. It achieved an R^2 score of 0.9988, meaning it explains nearly 99.9% of the variance in groundwater levels.

3.3 Challenges

The biggest hurdle was multicollinearity. Because ground elevation and reference point elevation are so closely related, it was a delicate balance to include enough information without confusing the models with redundant data.

4. Discussion

4.1 Interpretation

The high accuracy suggests that in this dataset, physical geography (elevation) is a dominant predictor of where the water table sits. The low RMSE (34.18) means our predictions are close enough to the actual measurements to be used for real-world planning.

4.2 Limitations & The Future

While successful, this model operates in a vacuum. It doesn't yet account for human factors like agricultural pumping or environmental factors like monthly rainfall. My suggestion for future research is to integrate precipitation data and pumping rates to see how the water table reacts to human and climatic stress.

FIGURES

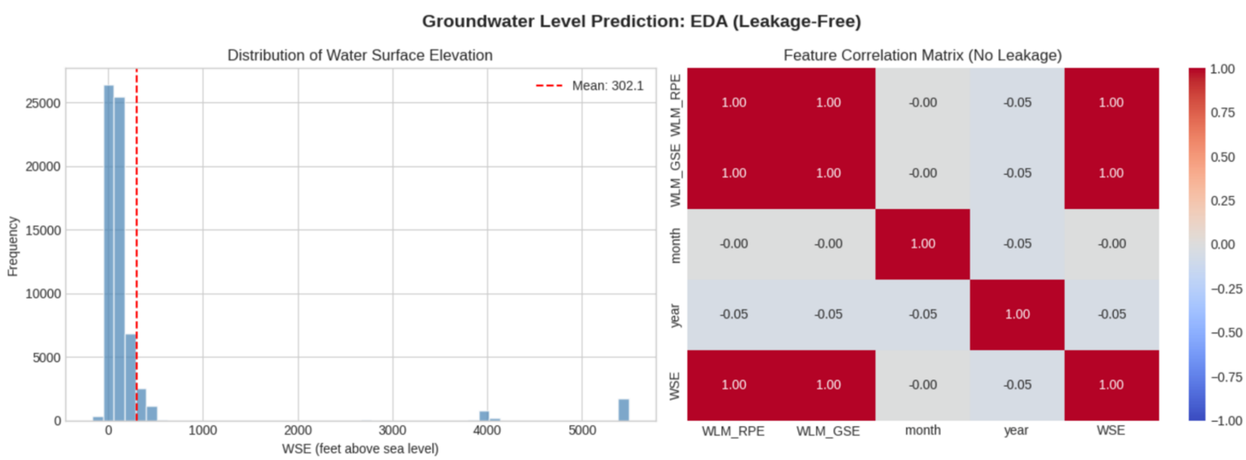


Fig. Distribution of water surface Elevation & Feature Correlation Matrix

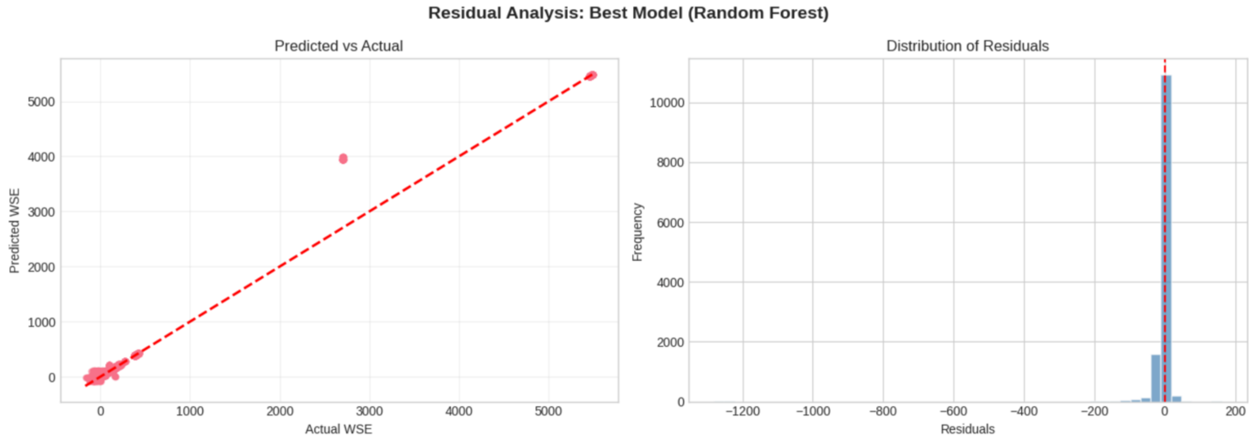
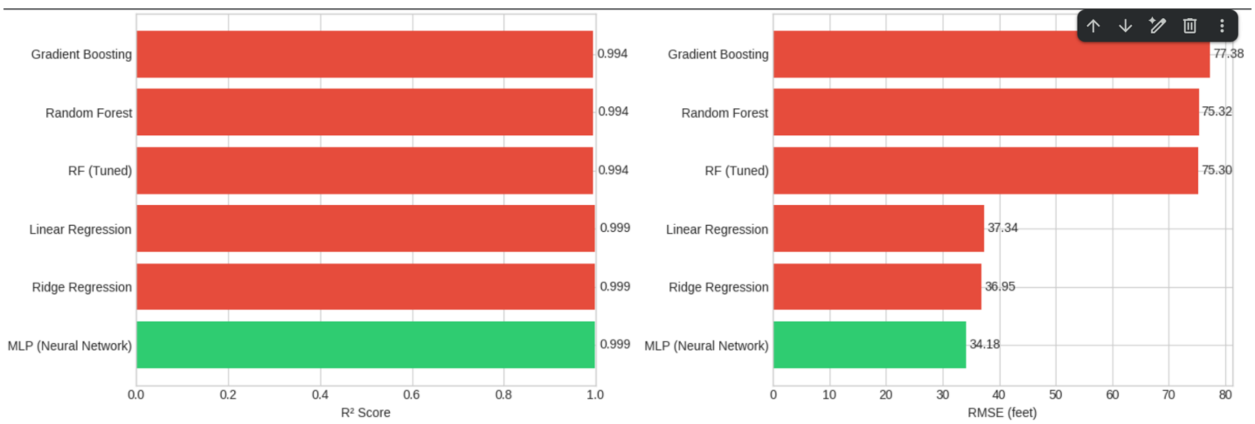


Fig. Predicted vs Actual & Distrubution of Residuals



5. References

- Scikit-learn: Machine Learning in Python, Pedregosa et al., JMLR 12, pp. 2825-2830, 2011.
- United Nations Sustainable Development Goals (Goal 6).